OGE-JEPA-TTC v6: Auditable Dense Event Prediction, Object Geometry, and a Local Garl-TTC Baseline

KriptaStudios Research Handoff
Reproducible local report from the e-jepa-ttc repository
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Abstract

This report describes version 6 of an event-camera time-to-collision research pipeline. The revision does not claim state of the art. It first reconstructs the historical Event-JEPA baseline exactly, separates that historical anchor from a matched architecture control, and implements staged tests for dense patch interaction, task-specific Attention Residuals, temporal delta memory, object geometry, ego-motion, and a local Garl-TTC replica. The historical checkpoint is reproduced byte-for-byte on its original cache and obtains 0.322892 s validation MAE, 0.584432 s RMSE, and 8.1554% mean relative error. FlowMimic-style global auxiliary losses are retained only as a negative result because they degrade multi-seed MAE.

The new components have passed unit and integration tests, but their small smokes are explicitly non-promotable. Version 6 therefore defines a staged screen, five-fold grouped cross-validation, and three-seed confirmation before any external evaluation. The official Benchmark-10 remains sealed. The main contribution at this stage is an auditable experimental basis rather than a performance claim.

1 Scope and claim boundary

The project estimates continuous TTC from asynchronous event streams and, conditionally, RGB and causal navigation. The long-term hypothesis is that preserving dense spatial evidence, isolating the target object, and embedding physical expansion in the output path can outperform regression from a pooled scene embedding. This hypothesis is not yet accepted by the evidence.

The data inventory is closed:

- EvTTC-32 is the only official TTC supervision source;
- eAP train-40 is reserved for later non-TTC pretraining;
- Benchmark-10 is an external sealed evaluation set;
- pseudo-TTC is never treated as ground truth.

Table 1: Exact historical BASE validation reproduction, seed 7.

Quantity	Value
Train windows	7,835
Validation windows	2,040
Downstream checkpoint	epoch 26/30
SSL checkpoint	epoch 6/30
MAE	0.322892 s
RMSE	0.584432 s
Mean relative error	8.1554%

Table 2: Historical multi-seed MAE. Lower is better.

Arm	MAE (s)	Change
BASE	0.332715	–
Global alignment	0.369604	+11.1%
Synthetic inverse-TTC	0.432909	+30.1%
Both losses	0.435869	+31.0%

2 Historical evidence

2.1 Exact BASE reconstruction

The audited baseline consumes 10 event channels and 11 causal auxiliary channels. Its EventTubeletTransformer uses dimension 192, depth 6, six heads, and patch size 16. Final dense tokens are mean-pooled and passed through LayerNorm, a 192-to-96 MLP, GELU, dropout 0.1, and a scalar log-TTC output.

Prediction arrays are byte-equivalent to the original stored predictions, and all audited metric differences are zero at tolerance 10^{-12}. This model is named `B0_HISTORICAL_BASE_EXACT`. It is not reused as if it had been trained under the new object-cache protocol.

2.2 Negative FlowMimic result

These losses are removed from the active architecture. Their summary remains versioned to preserve negative evidence.

3 Matched architecture protocol

The learned Core arms are:

- `A0_MATCHED_GLOBAL`, the object-cache global control;
- `A1_MATCHED_DENSE_BLOCK`, delayed pooling;
- `A2_MATCHED_DENSE_ATTNRES`, depth routing by task;
- `K1_OBJECT_KDA`, temporal delta memory.

The fairness audit requires identical sample IDs, backbone hash, common head hash, train and validation counts, optimizer, learning rate, weight decay, maximum epochs, log-TTC loss, and early stopping. Actual completed epochs may differ only through the shared stopping rule.

Core and Garl write separate stage roots:

```
artifacts/runs/evttc32_architecture_v4_<split>_<mode>/
  core/fold-<n>/matrix_summary.json
  garl/fold-<n>/matrix_summary.json
```

This prevents the previous summary collision while retaining individual run directories.

4 Dense spatial and temporal design

4.1 Patch Policy

Patches from one timestamp communicate bidirectionally before any temporal compression. Causality exists between time blocks, not between arbitrary patches inside a frame. Pooling before target localization is forbidden in the dense arm. This implements the dense representation principle motivated by Patch Policy [1].

4.2 Attention Residuals

Mask, motion, geometry, and risk tasks independently retrieve features from embedding and block outputs. Keys are RMS-normalized so a layer cannot dominate by scale; the mixed values remain the original features. A permanent greater-than-90% weight on one layer is treated as collapse.

4.3 Temporal delta memory

The KDA candidate follows the delta-recurrence principle described by Kimi K3 [4], but is restricted to the temporal axis. Spatial interaction is resolved first. KDA is optional and must show either a precision gain, a material memory reduction, or a longer effective horizon.

5 Local Garl-TTC baseline

Garl-TTC is implemented from the public source associated with the eAP work [3]. The audited local contract uses three timestamps at 100 ms cadence, RGB endpoints, two event intervals, 20 time planes per interval, and a common 128-by-128 endpoint square crop. RGB and event ResNet-50 branches can be trained for direct regression or learned height ratio (LHR). Late fusion initializes its branches from the corresponding unimodal LHR checkpoints. A foreground decoder produces four 256-by-256 logits during training and is absent from inference.

EvTTC does not expose the original eAP 3D visible-height target. The local adapter therefore supervises visible box height after the shared crop. Results must be reported as a local architectural replica, not official eAP parity.

6 Object geometry

For apparent scale s, inverse-TTC at the current endpoint is

$$q_t = \frac{s_t/s_{t-\Delta t} - 1}{\Delta t}. \tag{1}$$

Height, square-root area, and affine expansion all use the latest valid pair. The affine solver separates translation, radial expansion, and in-plane rotation through weighted least squares. Event contrast evaluates a bounded constant-approach warp only inside the object support. Auxiliary metadata channels are excluded from the event calculation.

The training-free bbox oracle reports each expert and a deterministic confidence mixture. It is an upper-bound diagnostic and not a bbox-free model.

7 Ego-motion

The EvTTC format states that integrated navigation attitude, position, and velocity live in a world coordinate system [2]. Local auditing shows velocity components ordered north, east, up and heading in degrees. The heading agrees with geodetic displacement. Velocity is transformed through the documented rigid chain

$$T_{\text{event}\leftarrow\text{nav}} = T_{\text{event}\leftarrow\text{RGB}} T_{\text{RGB}\leftarrow\text{LiDAR}} T_{\text{LiDAR}\leftarrow\text{nav}}. \tag{2}$$

The lever-arm velocity of the event-camera origin is included when heading changes.

For a past point with causal depth, full planar camera-pose compensation uses

$$P_c = R_{c\leftarrow p} P_p - \Delta C_c. \tag{3}$$

The boxes are reprojected with event-camera intrinsics. Ego closing contributes $q_{\text{ego}} = v_z/Z$. Translation is not observable from velocity alone: depth is required. Consequently, official EvTTC distance is allowed only for an oracle or teacher. A deployable arm must replace it with causal predicted depth and be re-evaluated.

Table 3: Evidence hierarchy.

Stage	Purpose
Smoke	Integration only, never promotion
Screen	304/80 windows, seed 7, up to 8 epochs
Grouped CV	Five folds, seed 7
Multi-seed	Seeds 7, 13, 21 for at most two finalists
External	Frozen Benchmark-10 inference

8 Experimental stages

The checkpoint score combines sequence-macro relative error, normalized RMSE, high-TTC error, and low-TTC safety error. Ties are resolved by worst sequence, RMSE, seed variance, and latency. A smoke with 38/10 windows and two epochs has already exercised all current paths, but its numerical ranking is not scientific evidence.

TargetQuery, predicted masks, high-resolution refinement, learned routing, bounded residual, and uncertainty remain behind a hard gate: bbox-GT geometry must first outperform BASE under the declared protocol.

9 Efficiency and storage

The consumer hardware target has approximately 12.8 GB VRAM and 32 GB RAM. Training uses BF16, pinned memory, persistent workers, prefetch, and gradient accumulation. The Garl ResNet-50 screen uses microbatch 8 with accumulation 16; the foreground arm uses microbatch 1 with accumulation 128. The geometric solver remains FP32.

No full eAP voxel cache, full-resolution SAM logits, all-layer DINO cache, or bulk TAR extraction is permitted. A run retains at most `best`, `last`, and `weights_only`. Core and Garl caches are compressed and stage-specific.

10 Limitations and next steps

No new component has a valid Screen or grouped-CV improvement yet. The Garl ResNet-50 topology is tested, but its real screen remains pending. Translation compensation is not deployable until depth is predicted. eAP is incomplete and has no official TTC labels. Benchmark-10 has not been opened.

The next actions are: execute Core Screen, execute Garl Screen, promote only arms passing their gates, run five-fold CV, repeat at most two finalists over three seeds, and freeze before any external inference. These restrictions preclude a current SOTA or safety claim.

References

[1] Anonymous. Patch policy: Efficient embodied control via dense visual representations. *arXiv preprint arXiv:2607.18236*, 2026.

[2] Jinghang Li et al. Evttc: An event camera dataset for time-to-collision estimation. *arXiv preprint arXiv:2412.05053*, 2024.

[3] Jinghang Li, Shichao Li, Qing Lian, Peiliang Li, Xiaozhi Chen, and Yi Zhou. Toward deep representation learning for event-enhanced visual autonomous perception: the eap dataset. *arXiv preprint arXiv:2603.16303*, 2026.

[4] Kimi Team. Kimi k3: Open frontier intelligence. *arXiv preprint arXiv:2607.24653*, 2026.